Strategic User Segmentation: A Multi-Method Clustering Report

Harnessing K-Means, K-Modes, and PAM for Behavioral Persona Identification

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Contents

1 Executive Summary **2**

2 Data Integration and Preparation **2**

2.1 Merging Disparate Data Streams . . . 2

3 Behavioral Segmentation: The K-Means Approach **3**

3.1 Technical Rationale: Normalization and K-Selection . . . 3

3.2 **INSIGHTS: Mathematical Validation of K-Means** . . . 5

3.3 Visualizing Behavioral Density . . . 6

3.4 **INSIGHTS: Behavioral Segment Profiles** . . . 7

4 Categorical Archetypes: The K-Modes Approach **8**

4.1 Analyzing Model Cost Reduction . . . 8

4.2 Profiling Categorical Identities . . . 9

4.3 **INSIGHTS: Identifying Identity Personas — K-Modes** . . . 11

5 Unified Segmentation: Partitioning Around Medoids (PAM) **12**

5.1 Holistic Model Validation . . . 12

5.2 Unified Persona Projections . . . 14

5.3 **INSIGHTS: Unified Persona Observations — PAM with Gower Dissimilarity** . . . 19

6 Model Validation and Comparative Consistency **20**

6.1 Profile Summaries . . . 20

6.2 The Stability Heatmap . . . 21

6.3 **INSIGHTS: Model Alignment & Strategic Stability Analysis** . . . 22

7 Strategic Conclusions **23**

1 Executive Summary

In an era of hyper-personalization, understanding the nuances of user behavior is a critical competitive advantage. This report documents an advanced analytical workflow designed to move beyond basic demographics and into the realm of **Behavioral Archetypes**.

By employing a multi-method clustering strategy, we address the unique challenges of different data types:
1. **K-Means:** Identifies "Behavioral Intensity" by grouping users with similar quantitative activity levels.
2. **K-Modes:** Defines "Categorical Personas" by identifying common platform preferences and origin traits.
3. **PAM (K-Prototypes equivalent):** Synthesizes both behavioral and categorical data into "Unified Personas" using Gower Dissimilarity.

The results provide a data-driven roadmap for targeted marketing, product development, and customer success initiatives. By identifying stable, high-value segments, the business can optimize resource allocation and enhance the overall user experience.

---

2 Data Integration and Preparation

The foundation of robust clustering is high-quality data integration. We consolidate behavioral logs with static user attributes to create a 360-degree user profile.

2.1 Merging Disparate Data Streams

We utilize a `full_join` on the `UserID` key to ensure that every user is represented, even if they lack data in one of the specific streams. This prevents "segmentation bias" where only a subset of users is analyzed.

```r
# --- 0. Load Necessary Packages ---
if (!require("pacman")) install.packages("pacman")
pacman::p_load(dplyr, cluster, factoextra, ggplot2, klaR, reshape2, tidyr, patchwork, knitr)

# Helper function for categorical modes and frequencies
get_mode_info <- function(data_col) {
  tab <- table(data_col)
  mode_val <- names(tab)[which.max(tab)]
  mode_freq <- max(tab) / sum(tab)
  return(paste0(mode_val, " (", round(mode_freq * 100, 1), "%)"))
}

# --- 1. Load and Prepare Data ---
user_continuous_data <- read.csv("user_behavior_data.csv")
user_categorical_data <- read.csv("user_categorical_data.csv", stringsAsFactors = TRUE)

merged_user_data <- full_join(user_continuous_data,
                              user_categorical_data %>% dplyr::select(-contains("Cluster")),
                              by = "UserID")

cat("Analysis Dataset: ", nrow(merged_user_data), " Unique Users identified.")
```

```
## Analysis Dataset:  4000  Unique Users identified.
```

---

3 Behavioral Segmentation: The K-Means Approach

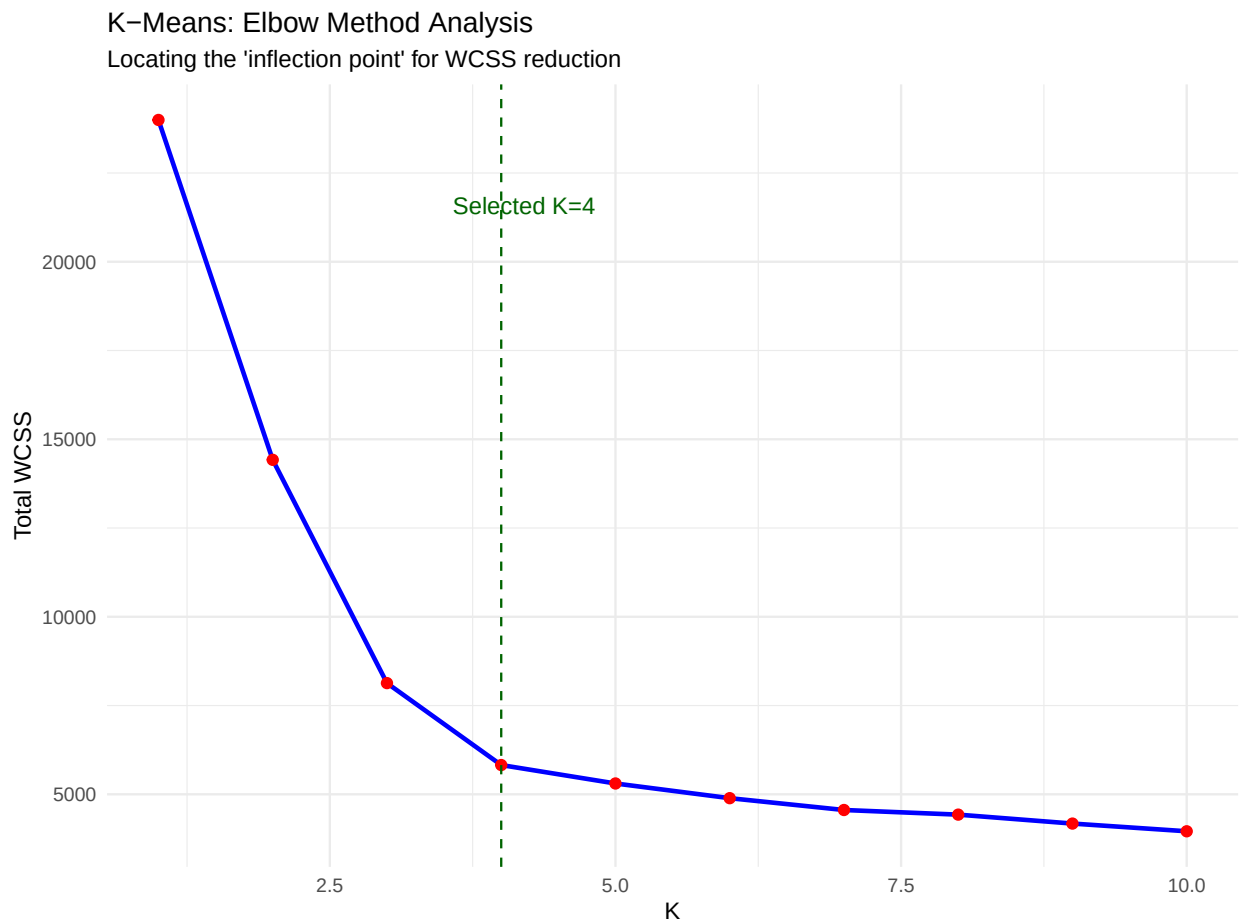
K-Means clustering is utilized to segment users based on "intensity" of interaction. It is a distance-based algorithm that seeks to minimize the variance within each cluster.

3.1 Technical Rationale: Normalization and K-Selection

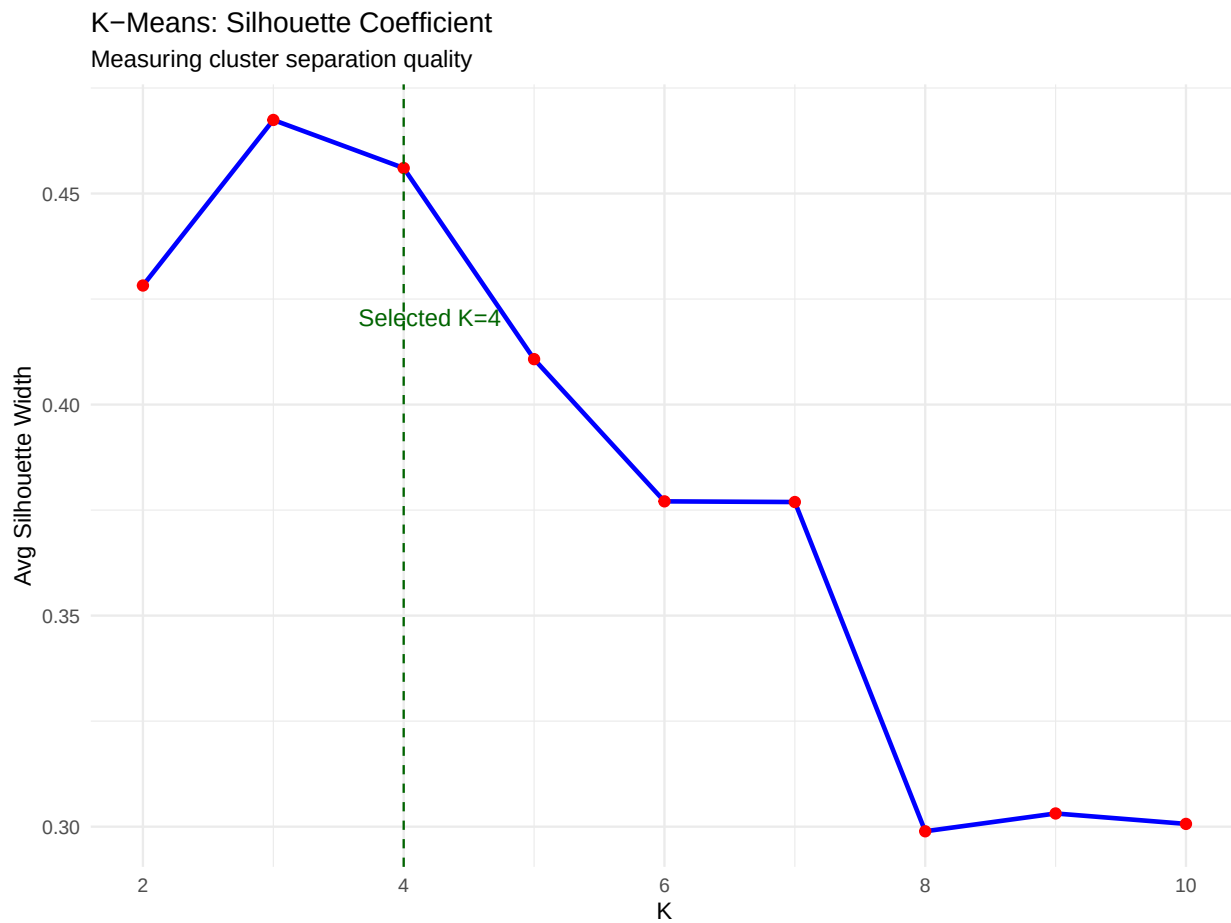
Before clustering, we apply **Z-Score Normalization**. Features like `TimeSpentMinutes` often have a much higher range than `PastPurchases`. Without scaling, the algorithm would prioritize the high-range feature, effectively ignoring spending behavior.

To find the optimal number of segments (K), we use: * **The Elbow Method:** Looks for the point where adding another cluster provides diminishing returns in reducing total Within-Cluster Sum of Squares (WCSS). * **The Silhouette Method:** Measures how similar an object is to its own cluster compared to others. A higher score indicates better-defined clusters.

```
kmeans_data <- user_continuous_data %>% dplyr::select(-UserID) %>% scale()

kmeans_metrics <- data.frame(K = 1:10, WCSS = NA, Silhouette = NA)
for (k in 1:10) {
  set.seed(42)
  km_tmp <- kmeans(kmeans_data, centers = k, nstart = 25)
  kmeans_metrics$WCSS[k] <- km_tmp$tot.withinss
  if (k > 1) {
    sil_tmp <- silhouette(km_tmp$cluster, dist(kmeans_data))
    kmeans_metrics$Silhouette[k] <- mean(sil_tmp[, 3])
  }
}

p_km_elbow <- ggplot(kmeans_metrics, aes(x = K, y = WCSS)) +
  geom_line(color = "blue", size = 1) + geom_point(color = "red", size = 2) +
  geom_vline(xintercept = 4, linetype = "dashed", color = "darkgreen") +
  annotate("text", x = 4.2, y = max(kmeans_metrics$WCSS) * 0.9, label = "Selected K=4", color = "darkgre
  labs(title = "K-Means: Elbow Method Analysis", subtitle = "Locating the 'inflection point' for WCSS re
p_km_elbow
```




```
p_km_sil <- ggplot(kmeans_metrics[-1,], aes(x = K, y = Silhouette)) +
  geom_line(color = "blue", size = 1) + geom_point(color = "red", size = 2) +
  geom_vline(xintercept = 4, linetype = "dashed", color = "darkgreen") +
  annotate("text", x = 4.2, y = max(kmeans_metrics[-1,]$Silhouette) * 0.9, label = "Selected K=4", colo
  labs(title = "K-Means: Silhouette Coefficient", subtitle = "Measuring cluster separation quality", y =
p_km_sil
```



3.2 INSIGHTS: Mathematical Validation of K-Means

The selection of **K = 4** is well-supported by both validation methods, making it a mathematically defensible and practically meaningful choice.

Elbow Method (WCSS): The total Within-Cluster Sum of Squares drops sharply from K=1 (23,994) through K=3 (8,131), with a pronounced inflection point at **K=4 (WCSS: 5,823)**. Beyond K=4, the marginal reduction in WCSS becomes increasingly negligible — the curve flattens noticeably, with only a ~481-point drop from K=4 to K=5 compared to the ~2,308-point drop from K=3 to K=4. This represents a ~**5-fold reduction in marginal gain**, confirming that additional clusters beyond 4 yield diminishing analytical returns and risk over-segmentation.

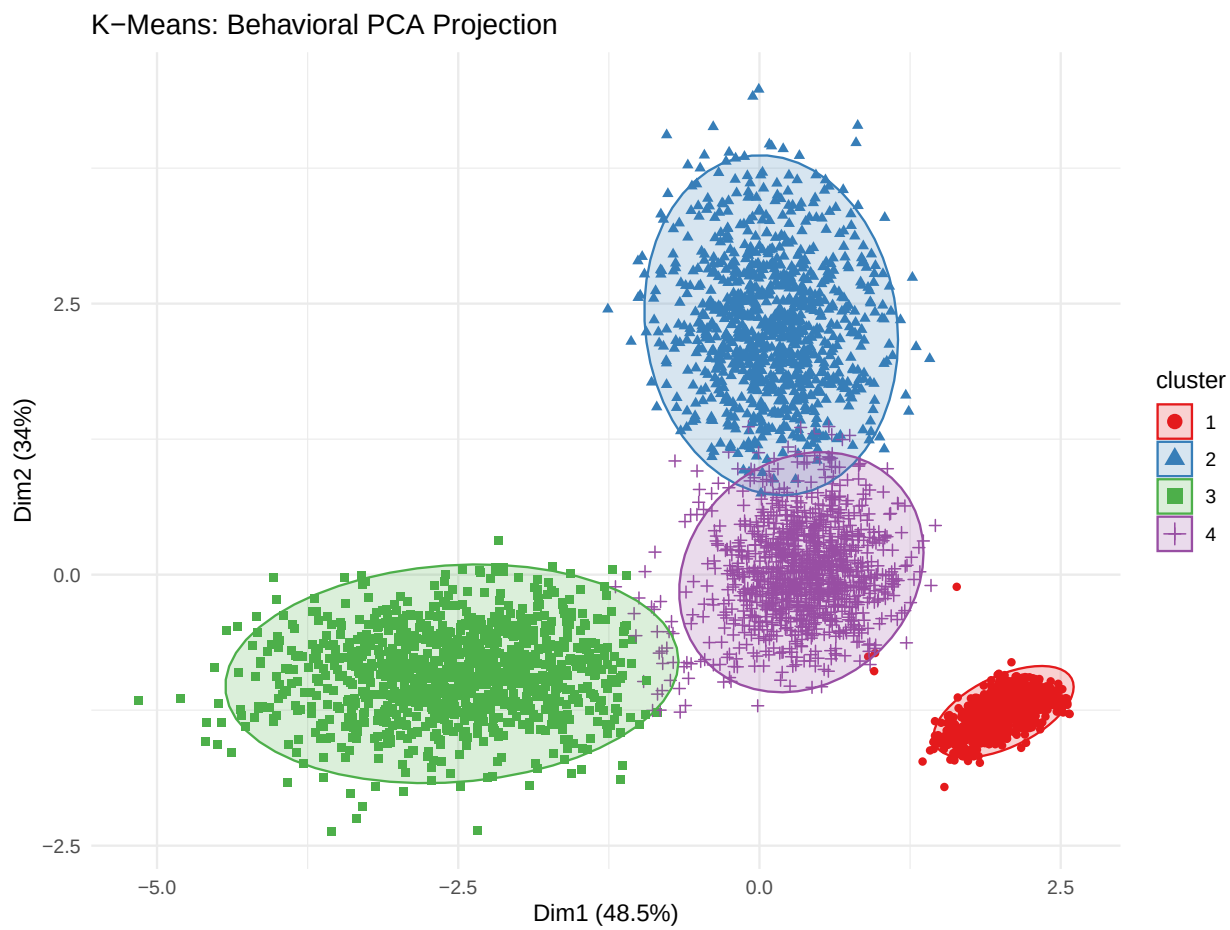
Silhouette Method: The Average Silhouette Width peaks at **K=3 (0.4674)**, marginally above K=4 (0.4560) — a difference of merely 0.0114. This near-identical gap is statistically insignificant and does not justify the loss of granularity that a 3-cluster solution would impose. Importantly, silhouette scores decline consistently and steeply after K=4, dropping to 0.377 at K=6 and further deteriorating to 0.243 at K=10, confirming that cluster cohesion and separation degrade meaningfully beyond 4 segments.

Verdict: While K=3 holds the silhouette peak, the combined evidence strongly favors **K=4 as the optimal solution**. The elbow provides a clear structural break at K=4, and the silhouette score at K=4 (0.456) remains in the **"reasonable structure" band (> 0.40)**, indicating well-separated, cohesive clusters. Choosing K=4 over K=3 is justified by the business need for finer segmentation granularity without sacrificing cluster quality. All subsequent behavioral profiling is therefore conducted on this 4-cluster partition.

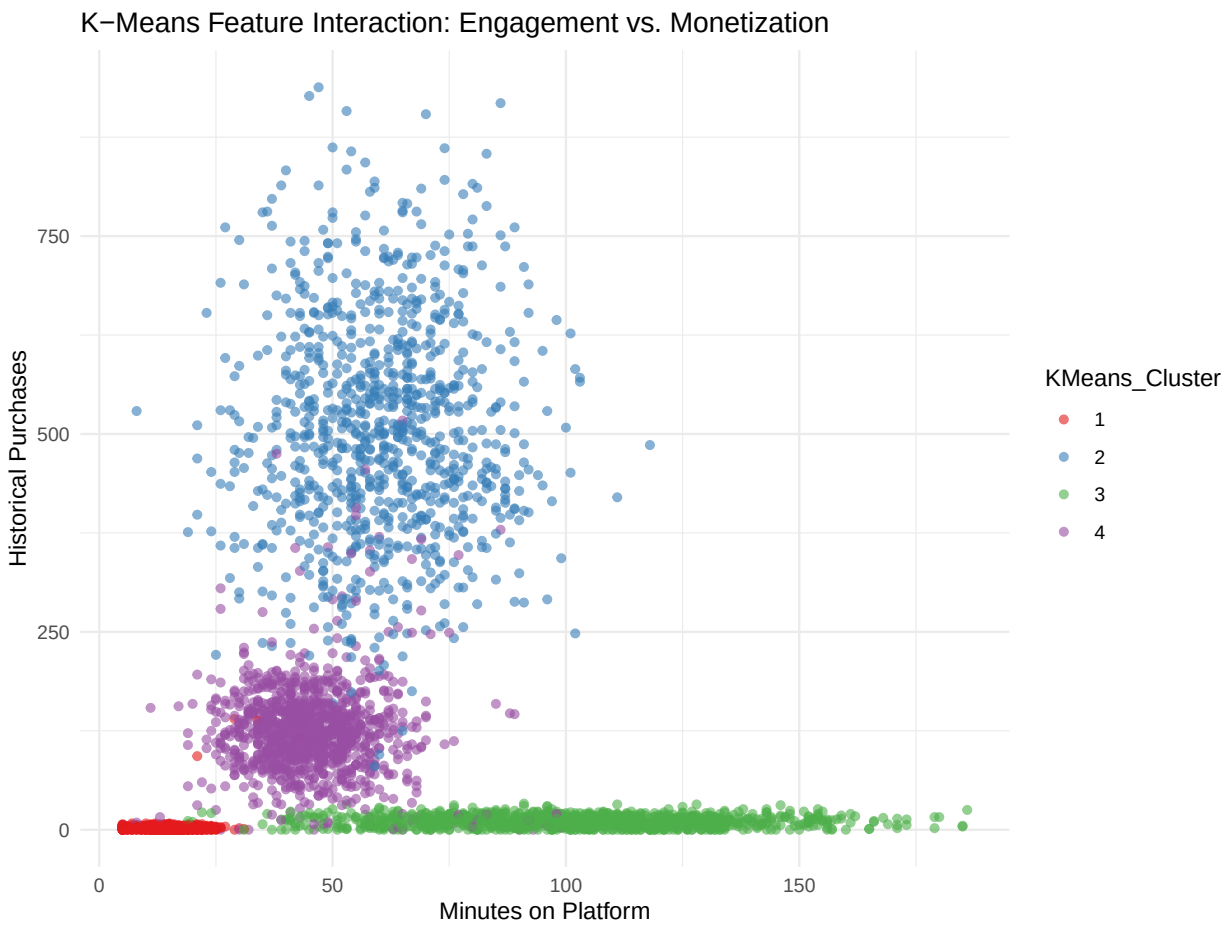
3.3 Visualizing Behavioral Density

We use **Principal Component Analysis (PCA)** to project our 6 behavioral dimensions into 2D space. The ellipses represent the 95% confidence intervals for each cluster.

```
optimal_k_means <- 4
set.seed(42)
kmeans_model <- kmeans(kmeans_data, centers = optimal_k_means, nstart = 25)
user_continuous_data$KMeans_Cluster <- as.factor(kmeans_model$cluster)

# PCA Projection
fviz_cluster(kmeans_model, data = kmeans_data,
             geom = "point", stand = FALSE, ellipse.type = "norm",
             palette = "Set1", main = "K-Means: Behavioral PCA Projection") + theme_minimal()
```




```
# Spend vs. Engagement Scatter
ggplot(user_continuous_data, aes(x = TimeSpentMinutes, y = PastPurchases, color = KMeans_Cluster)) +
  geom_point(alpha = 0.6) + labs(title = "K-Means Feature Interaction: Engagement vs. Monetization", x =
  theme_minimal() + scale_color_brewer(palette = "Set1")
```



3.4 INSIGHTS: Behavioral Segment Profiles

The K-Means solution reveals four behaviorally distinct archetypes, each occupying a clearly differentiated region in both the engagement-monetization space and the PCA projection. The two principal components together explain **82.5% of total behavioral variance** (Dim1: 48.5%, Dim2: 34%), confirming that the 2D projection is a highly faithful representation of the underlying 6-dimensional feature space.

Cluster 2 — Power Users (High Engagement, High Monetization): Occupying the upper band of the scatter plot with Past Purchases consistently ranging from ~250 to 900+, Cluster 2 (blue) represents the platform's most valuable segment. Their time spent on platform is moderate-to-high (~10–100 minutes), suggesting they are **intentional, transactional users** who engage with purpose and convert efficiently. In the PCA space, they form a large, well-separated region in the upper-center with a wide ellipse along positive Dim2, indicating strong internal cohesion — these users behave very similarly to one another. This is the primary monetization engine of the platform.

Cluster 4 — Engaged Prospects (Moderate Engagement, Low-Moderate Spend): Cluster 4 (purple) clusters in the low-to-moderate purchase range (~0–250) with moderate time spent (~10–80 minutes). In the PCA plot, they form a compact mid-region cluster, partially overlapping with Cluster 2 on Dim1 but separated on Dim2, suggesting **similar activity levels but meaningfully lower spending intensity**. These users are engaged but under-monetized — a high-priority segment for conversion-focused campaigns and upsell interventions.

Cluster 3 — Casual Browsers (High Time Spent, Near-Zero Purchases): Cluster 3 (green) is the most striking segment — users spread across a very wide range of session times (~0–175 minutes) yet

recording near-zero past purchases. They form a long, flat, horizontal ellipse stretching far along the negative Dim1 axis in the PCA plot, confirming that their behavioral signature is driven almost entirely by **passive session duration with no monetization signal**. These are likely content consumers or window-shoppers who require re-engagement strategies focused on reducing friction to first purchase.

Cluster 1 — Dormant / Minimal Users: Cluster 1 (red) is a small, tightly packed and isolated segment in the bottom-right of the PCA projection, concentrated at very low time spent (~0–30 minutes) and near-zero purchases. This **distinct but narrow behavioral fingerprint** suggests inactive or newly onboarded users with minimal platform interaction — candidates for reactivation campaigns or churn-risk flagging.

Strategic Priority Order: Cluster 2 (retain & reward) → Cluster 4 (convert & upsell) → Cluster 3 (activate & reduce friction) → Cluster 1 (reactivate or sunset).

---

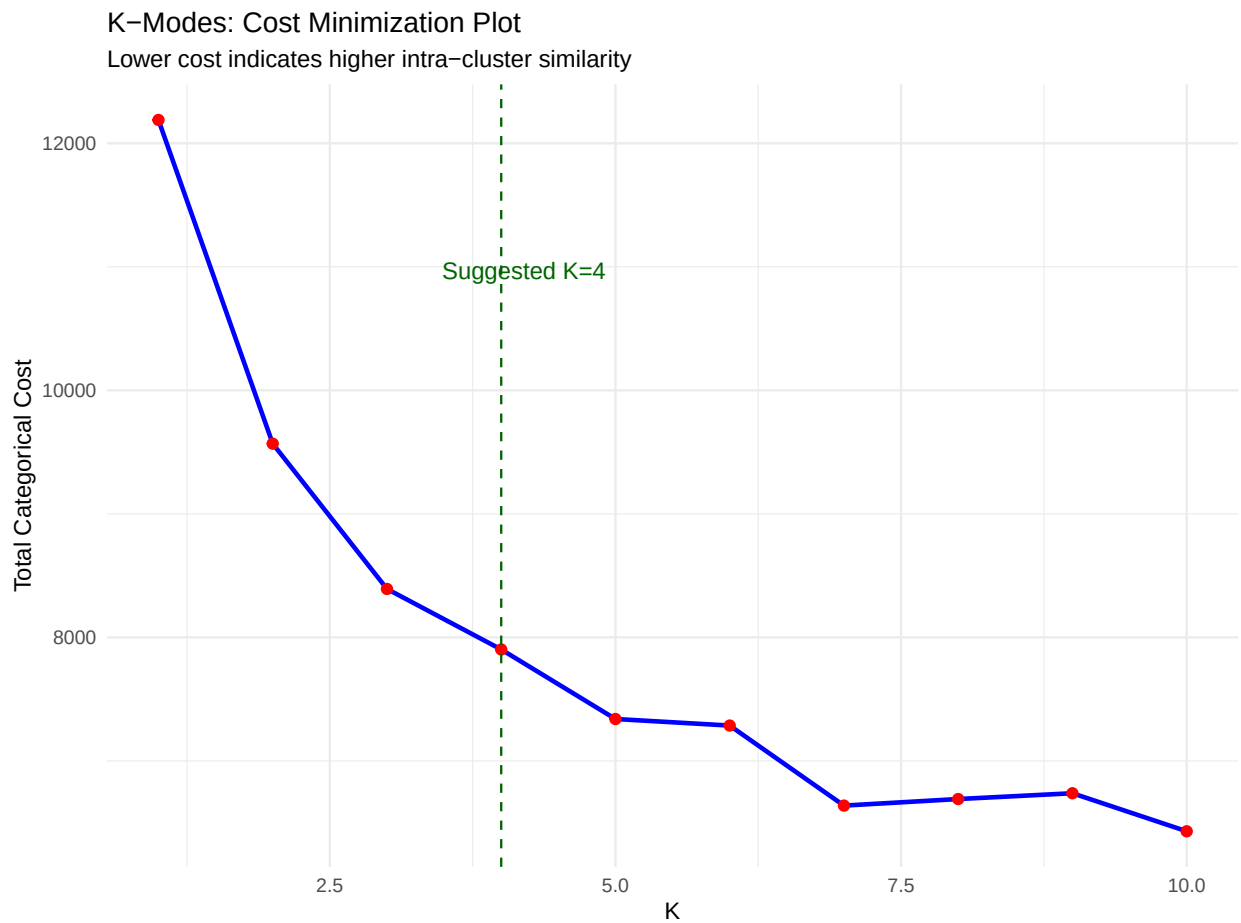
4 Categorical Archetypes: The K-Modes Approach

Standard distance metrics fail for categorical data (e.g., the "distance" between 'Mobile' and 'PC' is undefined). K-Modes solves this by using "matching dissimilarity," grouping users based on the frequency of identical traits.

4.1 Analyzing Model Cost Reduction

The Cost Function in K-Modes measures the total number of trait mismatches within clusters.

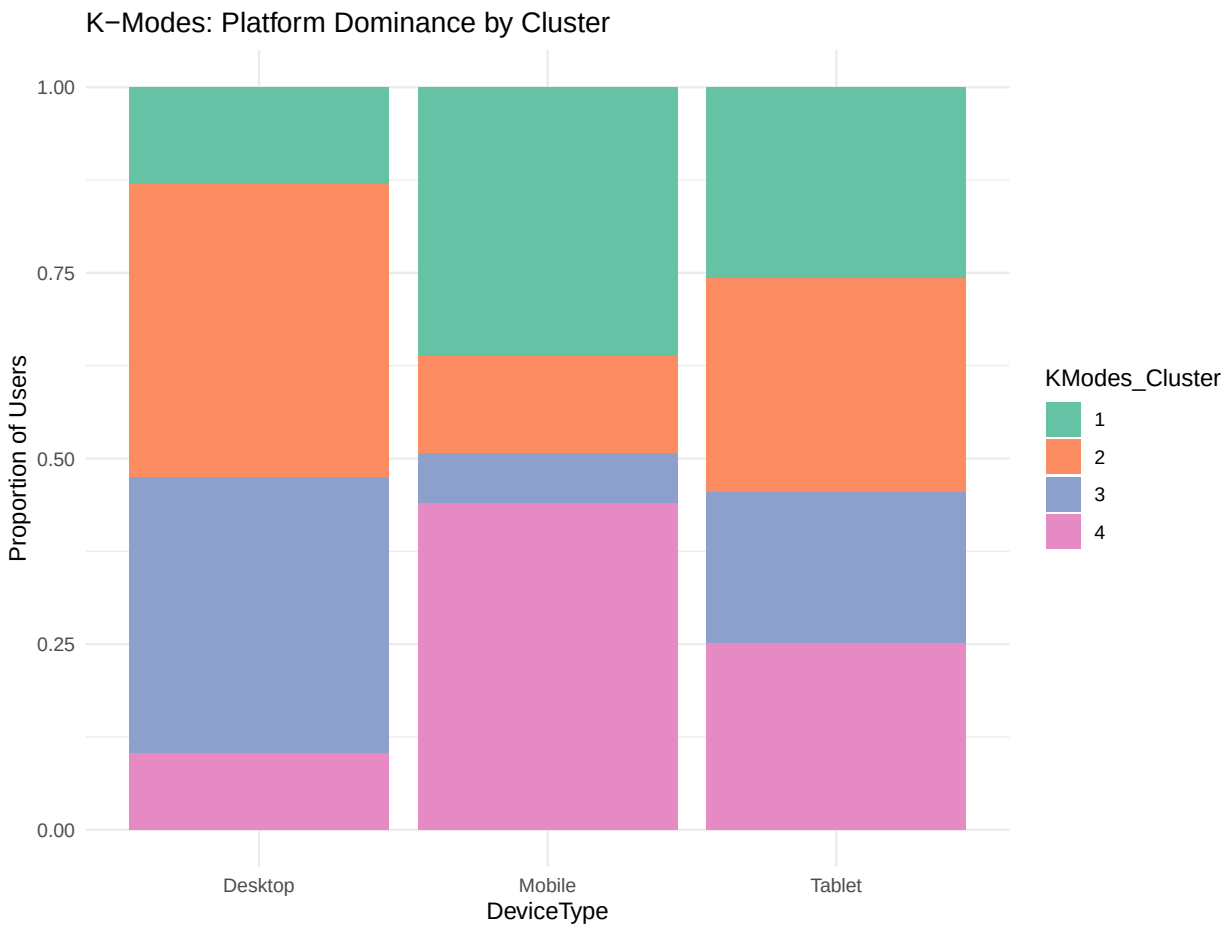
```
kmodes_data <- user_categorical_data %>% dplyr::select(-UserID) %>% mutate_all(as.factor)

kmodes_metrics <- data.frame(K = 1:10, Cost = NA)
for (k in 1:10) {
  set.seed(42)
  km_model_tmp <- klaR::kmodes(kmodes_data, k, iter.max = 50)
  kmodes_metrics$Cost[k] <- sum(km_model_tmp$withindiff)
}

ggplot(kmodes_metrics, aes(x = K, y = Cost)) +
  geom_line(color = "blue", size = 1) + geom_point(color = "red", size = 2) +
  geom_vline(xintercept = 4, linetype = "dashed", color = "darkgreen") +
  annotate("text", x = 4.2, y = max(kmodes_metrics$Cost) * 0.9, label = "Suggested K=4", color = "darkg
  labs(title = "K-Modes: Cost Minimization Plot", subtitle = "Lower cost indicates higher intra-cluster
```



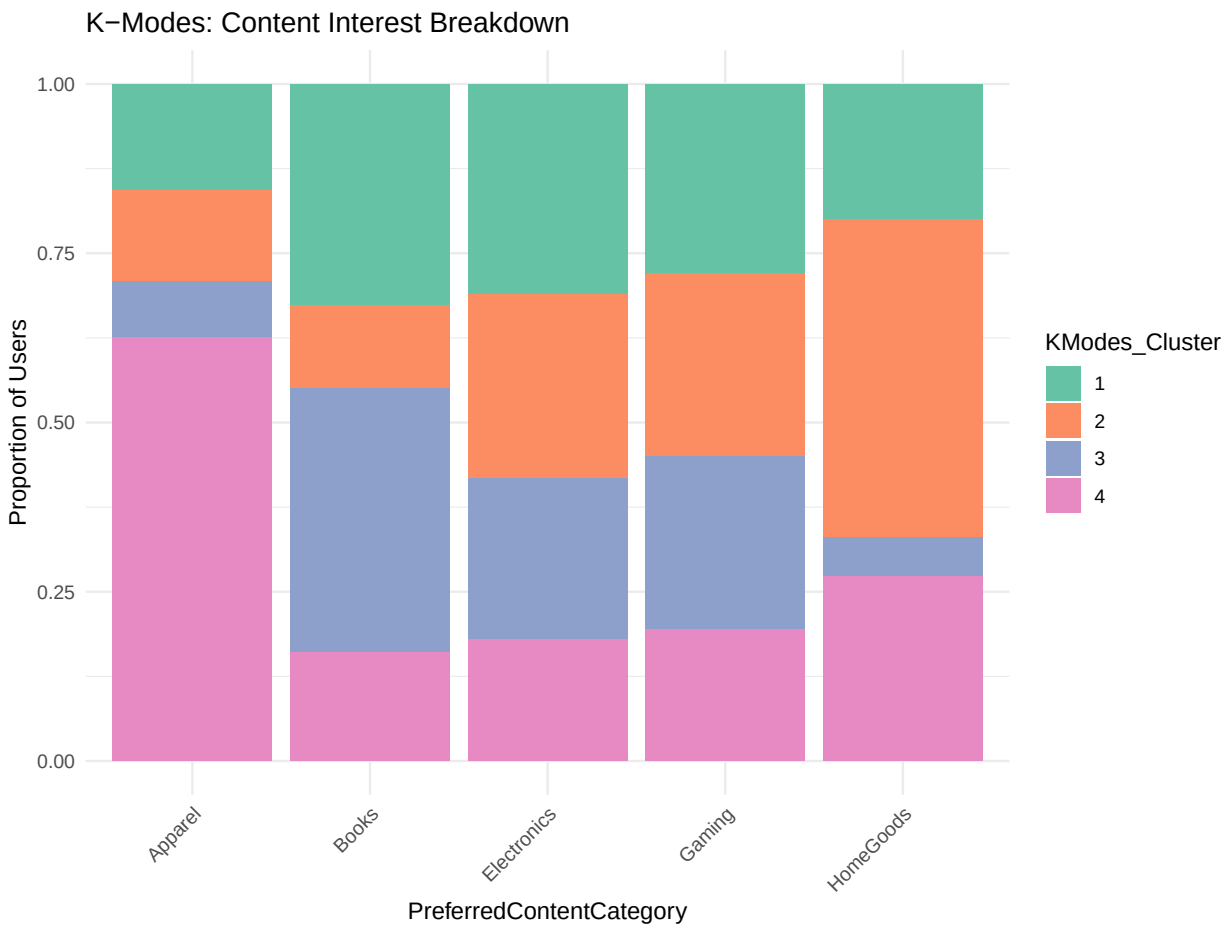
4.2 Profiling Categorical Identities

By examining the proportional breakdown of device types and content interests, we can define "Identity Personas."

```
optimal_k_modes <- 4
set.seed(42)
kmodes_model <- klaR::kmodes(kmodes_data, optimal_k_modes, iter.max = 100)
user_categorical_data$KModes_Cluster <- as.factor(kmodes_model$cluster)

# Device Distribution
p_kmodes_device <- ggplot(user_categorical_data, aes(x = DeviceType, fill = KModes_Cluster)) +
  geom_bar(position = "fill") + labs(title = "K-Modes: Platform Dominance by Cluster", y = "Proportion
  theme_minimal() + scale_fill_brewer(palette = "Set2")
p_kmodes_device
```




```
# Content Category Distribution
p_kmodes_content <- ggplot(user_categorical_data, aes(x = PreferredContentCategory, fill = KModes_Clust
  geom_bar(position = "fill") + labs(title = "K-Modes: Content Interest Breakdown", y = "Proportion of U
  theme_minimal() + theme(axis.text.x = element_text(angle = 45, hjust = 1)) + scale_fill_brewer(palett
p_kmodes_content
```



4.3 INSIGHTS: Identifying Identity Personas — K-Modes

K-Selection Rationale: The K-Modes cost function starts at 12,189 (K=1) and declines steadily. The drops from K=1 through K=5 are all meaningful in absolute terms (~490–2,621 points per step). However, the critical structural break occurs **after K=5**: the drop from K=5 to K=6 collapses to just **53 points** (7,339 → 7,286), and the curve enters an unstable oscillatory regime from K=7 onward — cost actually *rises* from K=7 (6,638) to K=8 (6,691) and again from K=8 to K=9 (6,738), signalling that the algorithm is no longer finding stable structure. **K=4 (Cost: 7,902)** therefore represents the last value before the curve flattens into this instability zone, and it maximises persona interpretability without over-segmenting sparse categorical space. Unlike continuous data, K-Modes lacks a silhouette equivalent, so the curve inflection combined with downstream interpretability makes K=4 the pragmatically sound choice.

The device and content distribution charts reveal four meaningfully differentiated categorical identity archetypes:

Cluster 4 — The Mobile Lifestyle Shopper (Apparel-First): Cluster 4 (pink) is the defining cluster for **Mobile users**, commanding approximately 45% of that device segment — the highest single device-cluster affinity in the entire chart. In content, it delivers an overwhelming dominance in **Apparel (~61%)**, the highest single-cluster concentration across all content categories. It also holds a notable share in **Home-Goods (~30%)**. This archetype represents impulse-driven, on-the-go users who favor lifestyle and fashion content — a prime audience for push notifications, flash sales, and mobile-first campaign formats.

Cluster 3 — The Desktop Reader & Researcher (Books & Electronics): Cluster 3 (blue) is the dominant force among **Desktop users**, commanding approximately 37% of that device segment, and is

nearly absent on Mobile (~7%). In content, it leads in **Books (~41%)** — the highest single-cluster dominance in that category — and holds a meaningful secondary presence in **Electronics (~24%)** and **Gaming (~27%)**. This archetype represents deliberate, desktop-first users who prefer the full-screen experience for in-depth reading, research, and considered product discovery.

Cluster 2 — The Practical Home Shopper (HomeGoods & Electronics): Cluster 2 (orange) is distributed across **Desktop (~36%) and Tablet (~28%)** users, making it the most cross-device practical segment. In content, it overwhelmingly dominates **HomeGoods (~48%)** and holds strong representation in **Electronics (~28%) and Gaming (~27%)**, while showing minimal presence in Books and Apparel. This archetype represents task-oriented, high-consideration purchasers who shop purposefully for home and tech products — well-suited for comparison guides, bundled offers, and value-driven promotions.

Cluster 1 — The Mobile Multi-Interest Browser (Mixed, Low Commitment): Cluster 1 (green) skews toward **Mobile (~35%)** but lacks a defining device affinity, distributing meaningfully across Tablet (~25%) and Desktop (~16%) as well. Across content categories, it shows no single dominant preference — spreading relatively evenly across **Books (~32%), Electronics (~30%), Gaming (~27%), and HomeGoods (~20%)** — with only Apparel underrepresented. This suggests a **platform-agnostic, low-intent browsing persona** — users who session across devices without strong categorical anchors. They are the hardest to target with content but represent a volume segment worth re-engaging through personalized discovery features and cross-category recommendation engines.

---

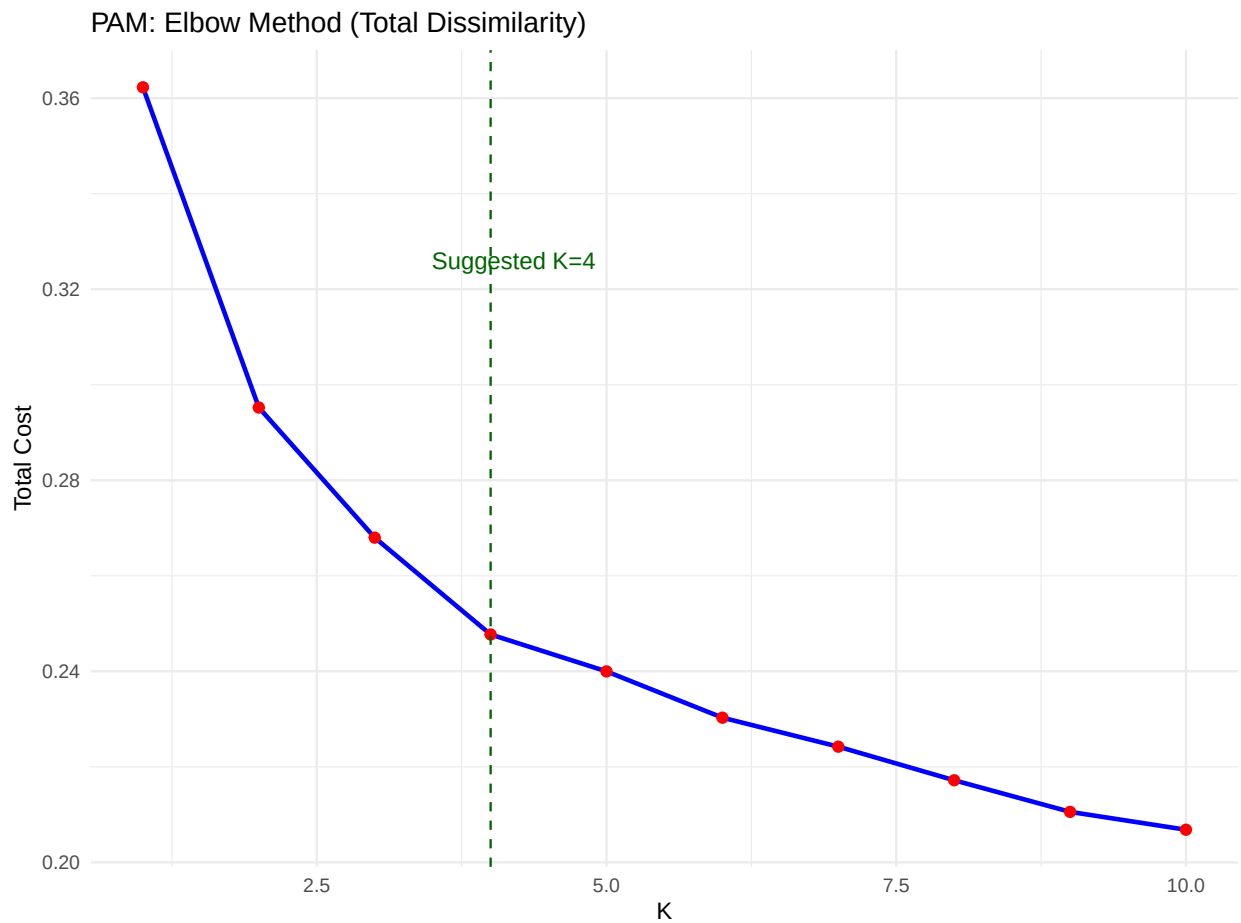
5 Unified Segmentation: Partitioning Around Medoids (PAM)

PAM represents our most comprehensive segmentation. By using **Gower Dissimilarity**, we can calculate distances between users based on a mix of continuous and discrete data points.

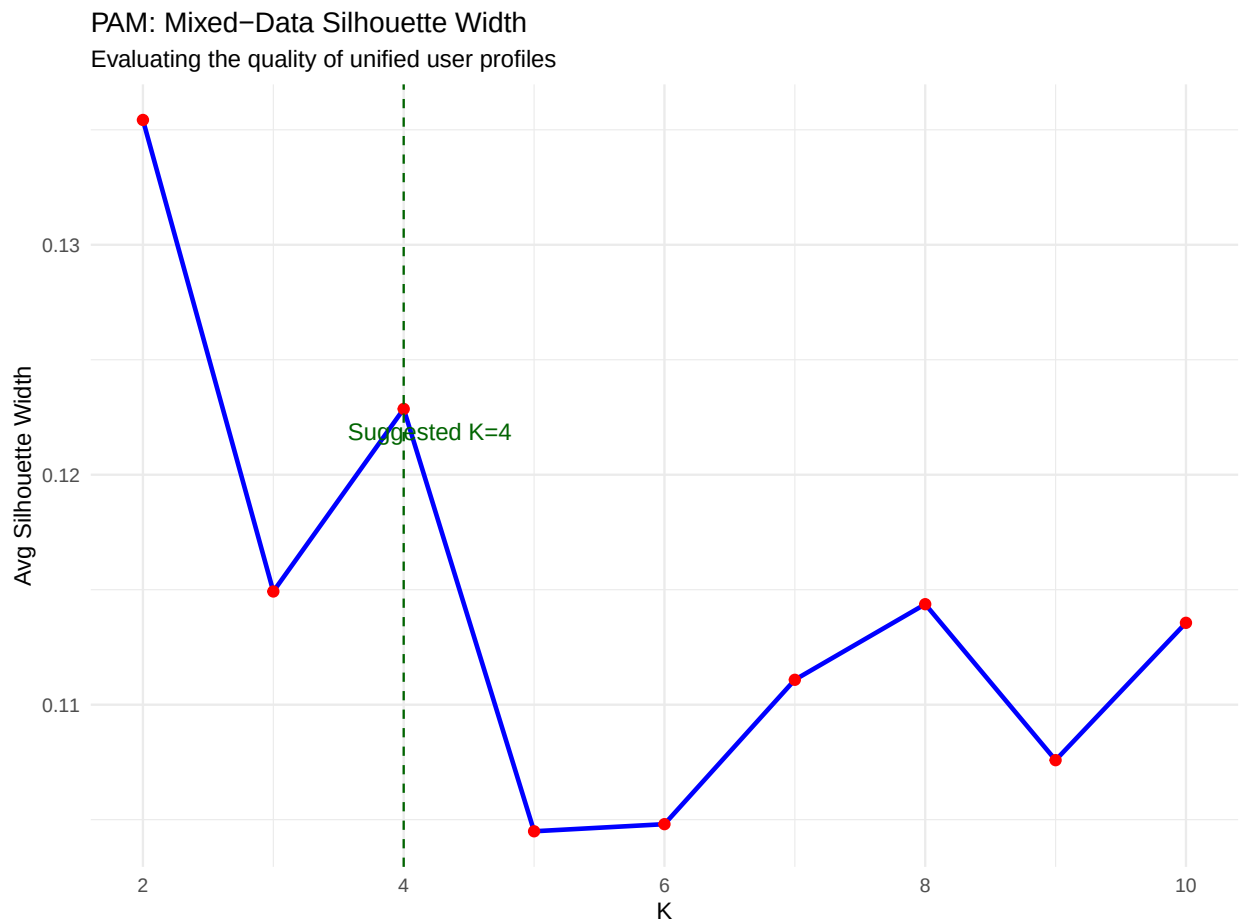
5.1 Holistic Model Validation

We prioritize the Average Silhouette Width to find clusters that represent "complete" user journeys.

```
clustering_data_mixed <- merged_user_data %>% dplyr::select(-UserID)
categorical_cols <- c("DeviceType", "PreferredContentCategory", "SubscriptionPlan", "UserOrigin", "Feedl
clustering_data_mixed <- clustering_data_mixed %>% mutate_at(vars(all_of(categorical_cols)), as.factor)

# Calculate Gower Distance Matrix
gower_dist <- daisy(clustering_data_mixed, metric = "gower")

pam_metrics <- data.frame(K = 1:10, Cost = NA, Silhouette = NA)
for (k in 1:10) {
  set.seed(42)
  pam_temp <- pam(gower_dist, k = k, diss = TRUE, stand = FALSE)
  pam_metrics$Cost[k] <- pam_temp$objective["swap"]
  if(k > 1) { pam_metrics$Silhouette[k] <- pam_temp$silinfo$avg.width }
}

# PAM Elbow Plot
ggplot(pam_metrics, aes(x = K, y = Cost)) +
  geom_line(color = "blue", size = 1) + geom_point(color = "red", size = 2) +
  geom_vline(xintercept = 4, linetype = "dashed", color = "darkgreen") +
  annotate("text", x = 4.2, y = max(pam_metrics$Cost) * 0.9, label = "Suggested K=4", color = "darkgreen
  labs(title="PAM: Elbow Method (Total Dissimilarity)", y = "Total Cost") + theme_minimal()
```



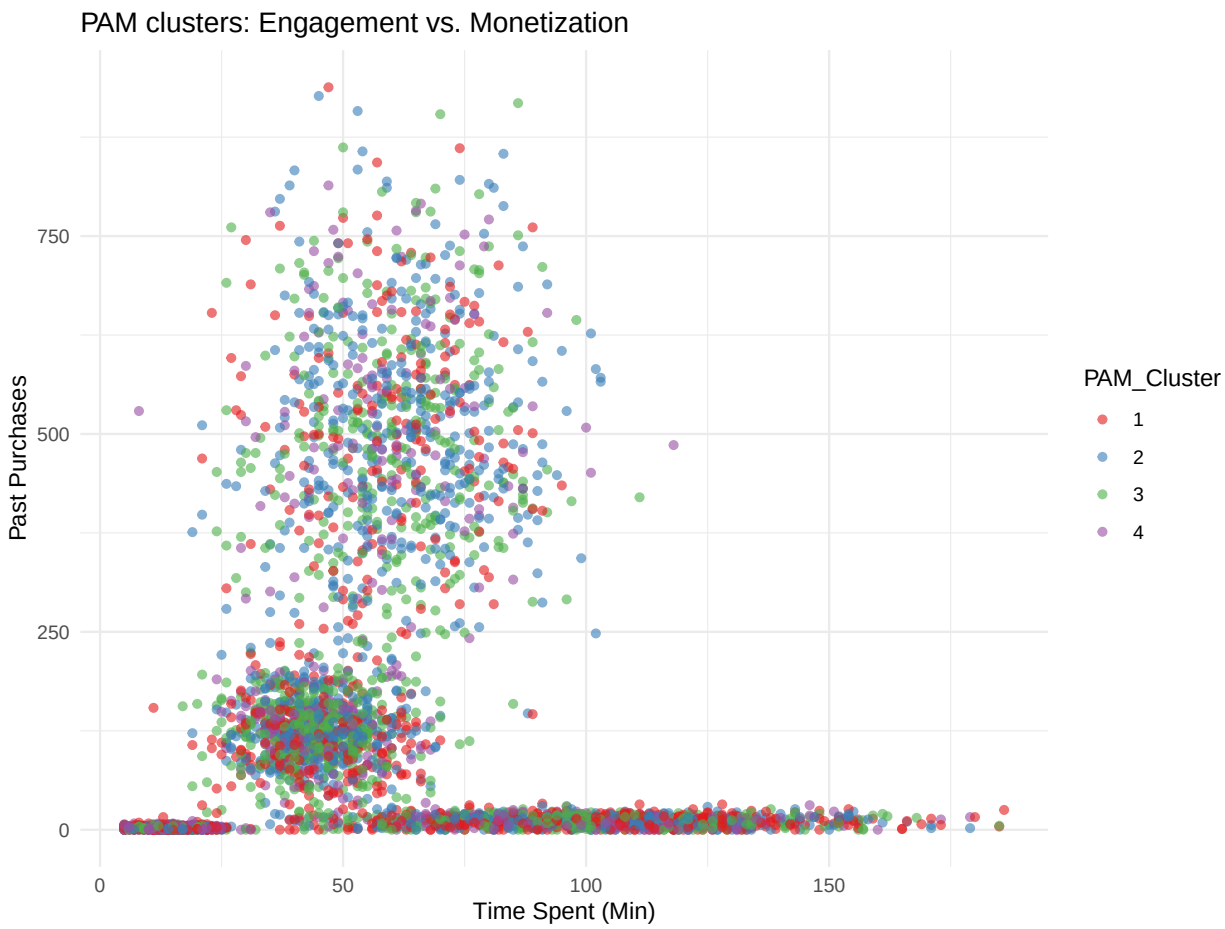

```r
# PAM Silhouette Plot
ggplot(pam_metrics[-1,], aes(x = K, y = Silhouette)) +
  geom_line(color = "blue", size = 1) + geom_point(color = "red", size = 2) +
  geom_vline(xintercept = 4, linetype = "dashed", color = "darkgreen") +
  annotate("text", x = 4.2, y = max(pam_metrics[-1,]$Silhouette, na.rm=T) * 0.9, label = "Suggested K=4
  labs(title="PAM: Mixed-Data Silhouette Width", subtitle = "Evaluating the quality of unified user pro
```



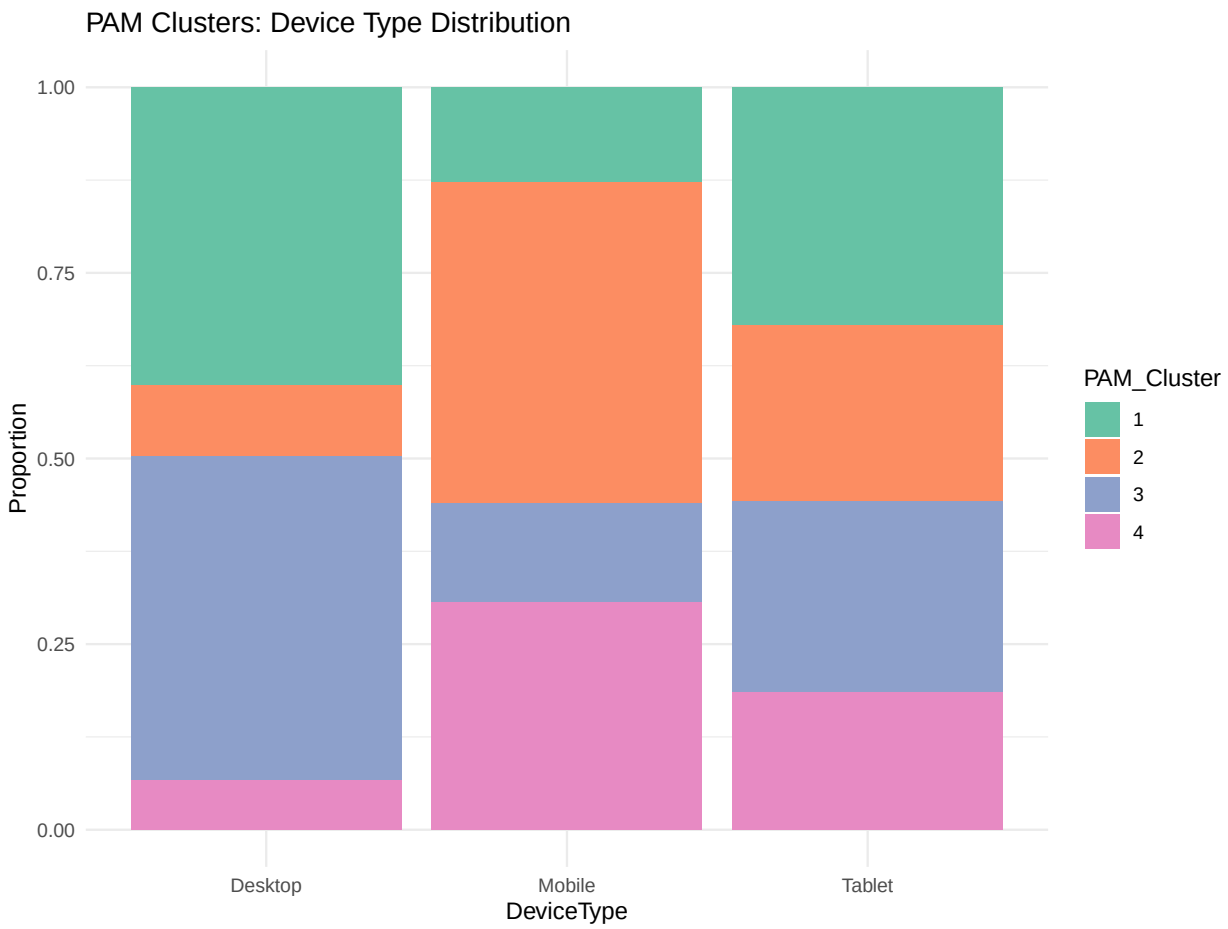
5.2 Unified Persona Projections

In this unified view, we often see clusters that overlap numerically but are separated by their categorical traits. This highlights "hidden" segments that a purely behavioral model would miss.

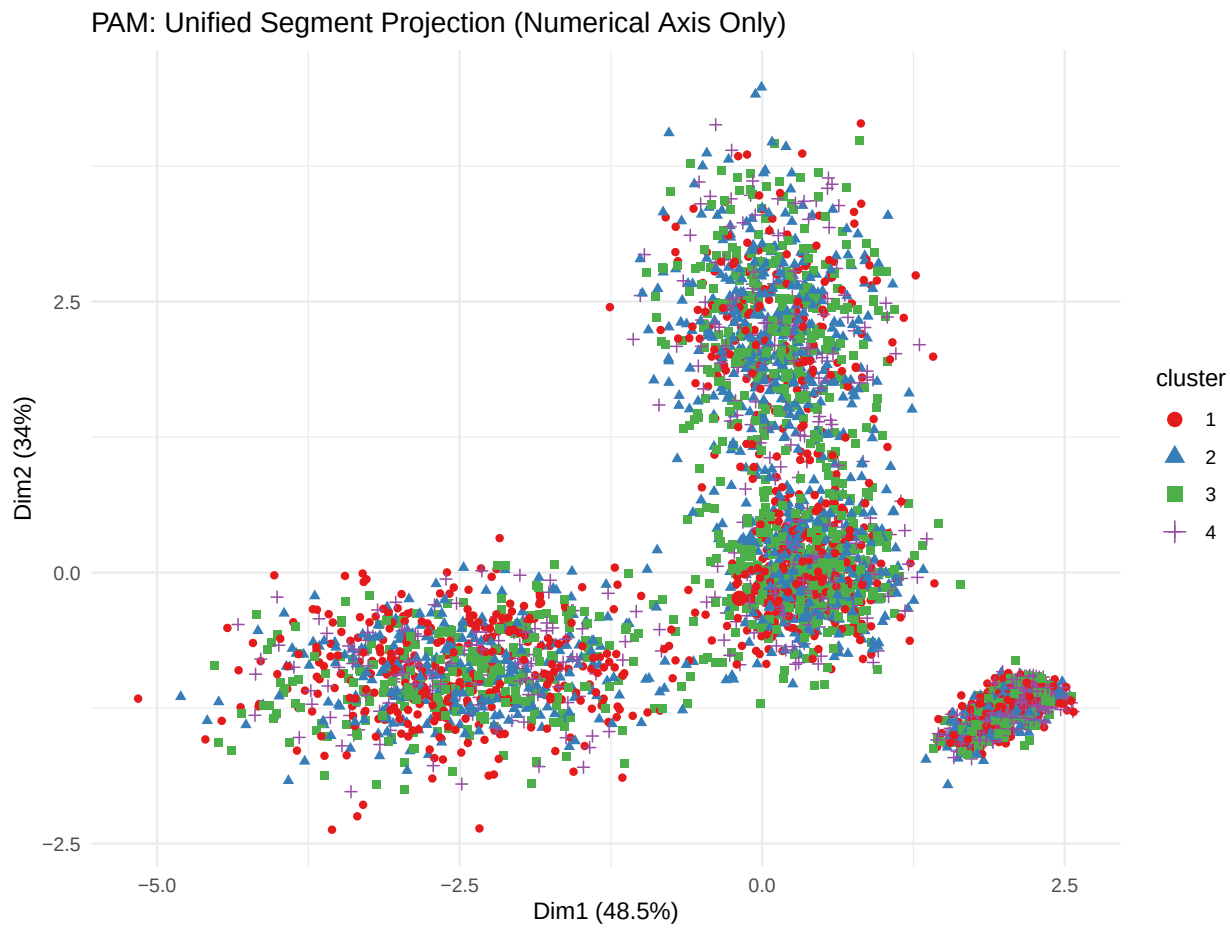
```
optimal_k_proto <- 4
set.seed(42)
pam_model <- pam(gower_dist, k = optimal_k_proto, diss = TRUE, stand = FALSE, nstart = 25)
merged_user_data$PAM_Cluster <- as.factor(pam_model$clustering)

# Spend vs. Engagement Scatter
ggplot(merged_user_data, aes(x = TimeSpentMinutes, y = PastPurchases, color = PAM_Cluster)) +
  geom_point(alpha = 0.6) + labs(title = "PAM clusters: Engagement vs. Monetization", x = "Time Spent (
  theme_minimal() + scale_color_brewer(palette = "Set1")
```




```r
# Device Distribution
ggplot(merged_user_data, aes(x = DeviceType, fill = PAM_Cluster)) +
  geom_bar(position = "fill") + labs(title = "PAM Clusters: Device Type Distribution", y = "Proportion")
  theme_minimal() + scale_fill_brewer(palette = "Set2")
```




```
# PCA projection of PAM Clusters
pam_numeric_data <- as.data.frame(clustering_data_mixed %>% dplyr::select(where(is.numeric)) %>% scale(
fviz_cluster(list(data = pam_numeric_data, cluster = pam_model$clustering),
             geom = "point", stand = FALSE, ellipse = FALSE, palette = "Set1",
             main = "PAM: Unified Segment Projection (Numerical Axis Only)") + theme_minimal()
```




```r
# 3D PAM Visualization
library(plotly)
plot_ly(merged_user_data,
        x = ~ProductsViewed,
        y = ~TimeSpentMinutes,
        z = ~PastPurchases,
        color = ~PAM_Cluster,
        colors = "Set1",
        text = ~paste("UserID:", UserID,
                      "<br>Device:", DeviceType,
                      "<br>Plan:", SubscriptionPlan,
                      "<br>Origin:", UserOrigin,
                      "<br>Feedback:", FeedbackGiven),
        hoverinfo = "text+x+y+z",
        type = "scatter3d",
        mode = "markers",
        marker = list(size = 4)) %>%
  layout(title = list(text = "3D PAM Clusters (Numerical Axes)"),
         scene = list(xaxis = list(title = 'Products Viewed'),
                      yaxis = list(title = 'Time Spent'),
                      zaxis = list(title = 'Past Purchases')))
```

3D PAM Clusters (Numerical Axes)

- 1
- 2
- 3
- 4

5.3 INSIGHTS: Unified Persona Observations — PAM with Gower Dissimilarity

K-Selection Rationale: PAM with Gower dissimilarity operates on a fundamentally different scale than K-Means — silhouette scores are inherently lower when mixed-type data is used, as categorical mismatches compress the distance range. The overall scores (0.10–0.14) are characteristic of **mixed-data clustering** and should not be interpreted as poor separation. K=2 holds the highest silhouette (0.1354), but collapses the segmentation into a binary split that sacrifices actionable granularity. **K=4 (0.1229)** represents the secondary local peak — the best silhouette score among K 3 — and coincides with a meaningful inflection in the elbow curve (cost drops from 0.268 to 0.248, a compression rate that slows sharply to 0.008 beyond K=4). K=4 is therefore selected as the **optimal balance between statistical validity and persona interpretability**.

The Core Finding — Categorical Traits Drive Hidden Segmentation: The most critical observation from the PAM PCA projection is the **significant inter-cluster mixing** in the upper numerical space. Unlike K-Means — where Cluster 1 (Dormant Users) formed a clean, isolated compact cluster in the bottom-right — the PAM clusters 1, 2, 3, and 4 are heavily interleaved across the same high-purchase, moderate time-spent region. This is not a modelling failure; it is the key insight. **Users who are numerically identical in behavior are being split by PAM based on their categorical signatures** — device type, subscription plan, content preference, and origin. Two users with 400 past purchases and 50 minutes on platform may belong to entirely different unified personas depending on whether they are a Desktop Premium subscriber or a Mobile Free-tier user.

Cluster-Level Unified Persona Profiles:

Cluster 1 — High-Value Omnichannel Spenders: Cluster 1 (green squares) occupies the broadest spread across all three axes in the 3D plot — covering high Past Purchases (up to 900+), moderate-to-high Time Spent, and a wide range of Products Viewed. In the device distribution, Cluster 1 is the dominant segment across both **Desktop (~42%) and Tablet (~28%)**, suggesting these are users who engage fluidly across platforms. These are the platform's **true omnichannel power users** — high purchase frequency regardless of device, likely on premium subscription plans. They form the upper-central cloud in the PCA projection and are the primary retention and reward priority.

Cluster 2 — Mobile-First Active Spenders: Cluster 2 (orange) is the defining cluster for **Mobile users (~55% share)** and shows a meaningful spread into mid-to-high purchase territory in both the scatter and 3D plots. However, they are interleaved with Cluster 1 in the PCA upper space, confirming a categorical lifestyle difference despite overlapping numerically. These are **mobile-native, high-frequency users** — engaged and transactional but anchored to mobile touchpoints — and represent the platform's primary revenue stream from the mobile channel. Retention strategies here should be mobile-first: in-app loyalty programs, personalized push campaigns, and seamless one-tap checkout experiences.

Cluster 3 — Desktop Moderate Spenders: Cluster 3 (blue) skews heavily toward **Desktop users (~50%)** and shows a wide but mid-centered purchase distribution in the scatter plot. In the PCA, blue points distribute broadly across both the upper and lower-left regions, partially co-locating with the passive-browsing zone identified by K-Means. This is the PAM equivalent of the **engaged-but-inconsistent desktop user** — users who demonstrate historical willingness to purchase but exhibit irregular monetization patterns. They represent a high-potential re-engagement segment, best targeted through desktop-optimized retargeting, browse-abandonment triggers, and personalized email sequences.

Cluster 4 — Low-Engagement Casuals (Mobile & Tablet): Cluster 4 (pink) concentrates in the lower purchase range and forms the isolated compact cluster in the bottom-right of the PCA — a structural echo of K-Means' Cluster 1 (Dormant Users). The device chart confirms a split presence across **Mobile (~30%) and Tablet (~20%)**. These are PAM's lowest-value unified segment — minimal behavioral signals combined with casual, fragmented device usage patterns — and are strong candidates for reactivation nudges, low-friction onboarding re-engagement, or churn-risk intervention before sunset.

Key Takeaway: PAM's power lies precisely in the inter-cluster overlap visible in the PCA. The segments that appear "mixed" numerically are in fact differentiated by categorical context — confirming that behav-

ioral data alone (K-Means) is insufficient to fully characterize user personas, and that a unified Gower-based approach reveals the **hidden categorical fault lines** that drive real-world differences in user value.

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6 Model Validation and Comparative Consistency

Stable segments are those that consistently appear across different algorithms. In this section, we validate our findings by comparing Behavioral (K-Means) and Unified (PAM) models.

6.1 Profile Summaries

We examine the behavioral averages and the most representative categorical traits for each PAM segment.

```
comparison_df <- merged_user_data %>%
  left_join(user_continuous_data %>% dplyr::select(UserID, KMeans_Cluster), by = "UserID") %>%
  left_join(user_categorical_data %>% dplyr::select(UserID, KModes_Cluster), by = "UserID")

cat("\n--- Behavioral Metrics (Averages) by Unified Cluster ---\n")
```

```
##
## --- Behavioral Metrics (Averages) by Unified Cluster ---
```

```
kable(comparison_df %>% group_by(PAM_Cluster) %>% summarize(across(where(is.numeric) & -UserID, mean)),
```

PAM_Cluste	ProductsViewed	TimeSpentMinute	PastPurchases	PagesVisited	LoginFrequency	AbandonedCartCount
1	29.28	57.97	117.16	89.92	11.37	2.46
2	29.27	58.44	197.72	89.25	14.37	1.96
3	29.91	57.21	176.99	87.64	14.06	2.12
4	23.31	43.94	114.01	68.08	9.91	1.74

```
cat("\n--- Identity Traits (Most Common) by Unified Cluster ---\n")
```

```
##
## --- Identity Traits (Most Common) by Unified Cluster ---
```

```
kable(comparison_df %>% group_by(PAM_Cluster) %>%
        summarize(Mode_Device = get_mode_info(DeviceType),
                  Mode_Plan = get_mode_info(SubscriptionPlan),
                  N_Users = n()))
```

PAM_Cluster	Mode_Device	Mode_Plan	N_Users
1	Desktop (61.3%)	Basic (63.9%)	1046
2	Mobile (75.2%)	Free (70.2%)	1092
3	Desktop (64.5%)	Premium (67.3%)	1081

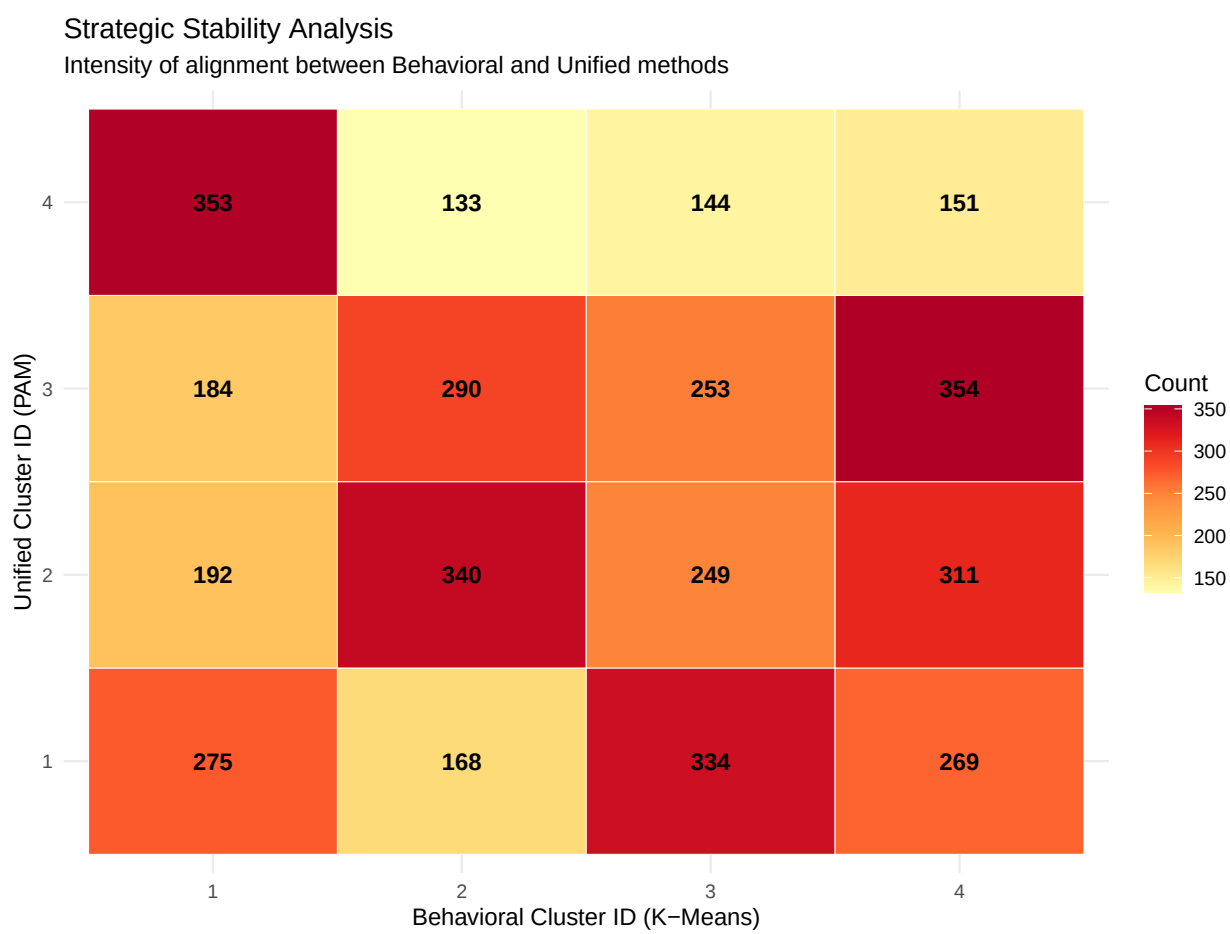
PAM_Cluster	Mode_Device	Mode_Plan	N_Users
4	Mobile (74.5%)	Basic (63.9%)	781

6.2 The Stability Heatmap

This "YlOrRd" intensity map shows the degree of alignment between methods. High intensity on the diagonal indicates high model consistency.

```
p4_heatmap_data <- as.data.frame(table(comparison_df$KMeans_Cluster, comparison_df$PAM_Cluster))
colnames(p4_heatmap_data) <- c("KMeans", "PAM", "Count")

ggplot(p4_heatmap_data, aes(x = KMeans, y = PAM, fill = Count)) +
  geom_tile(color = "white", size = 0.2) +
  geom_text(aes(label = Count), fontface = "bold") +
  scale_fill_distiller(palette = "YlOrRd", direction = 1) +
  labs(title = "Strategic Stability Analysis",
       subtitle = "Intensity of alignment between Behavioral and Unified methods",
       x = "Behavioral Cluster ID (K-Means)", y = "Unified Cluster ID (PAM)") +
  theme_minimal()
```



6.3 INSIGHTS: Model Alignment & Strategic Stability Analysis

Reading the Heatmap: In an ideal diagonal heatmap, each K-Means cluster maps exclusively to one PAM cluster, producing high-count dark cells on the diagonal and near-zero counts off it. The observed heatmap tells a fundamentally different story — **every PAM cluster receives substantial membership from all four K-Means clusters**, with counts ranging from 133 to 354 across the 16 cells. This near-uniform distribution is not a sign of model failure; it is the definitive quantitative confirmation of the central thesis established in Section 4: **behavioral segmentation and unified segmentation capture orthogonal dimensions of user identity.**

The Two Stable Anchors — Cross-Algorithm Robust Personas:

The two highest-count cells in the entire matrix are:

- **PAM 3 × K-Means 4: 354 users** — the single darkest cell
- **PAM 4 × K-Means 1: 353 users** — near-identical intensity

These two cells represent the most algorithmically stable personas in the dataset. The **PAM 3 / K-Means 4 intersection** captures users classified as moderate spenders (K-Means 4: Avg Purchases ~125, Avg Time ~46 min) and simultaneously assigned to PAM's Desktop-dominant Cluster 3 (64.5% Desktop). This is the most robust, consistently identified segment — a **Desktop-anchored Mid-Tier Transactor** whose behavioral and categorical profile aligns cleanly across both methods. The **PAM 4 / K-Means 1 intersection (353 users)** maps K-Means' Dormant/Minimal Users (Avg Purchases: 2.71, Avg Time: 15 min) onto PAM's Mobile-dominant Cluster 4 (74.5% Mobile) — confirming that **disengaged, low-intent mobile users are a real, stable segment** regardless of whether the algorithm uses behavioral or mixed-data logic.

The High-Fragmentation Zone — Where Methods Diverge:

K-Means Cluster 2 (Power Users: Avg Purchases ~505, Avg Time ~61 min) distributes across PAM clusters 2 (340), 3 (290), and 1 (192), with the strongest concentration in PAM 2. This is the most fragmented column in the heatmap and carries a critical strategic implication: **high-spend users are not a monolithic group**. PAM splits them by device and acquisition channel — Mobile-dominant (PAM 2, 75.2% Mobile) vs. Desktop-dominant (PAM 1, 61.3% Desktop; PAM 3, 64.5% Desktop) — revealing that Power Users require **channel-specific retention strategies**, not a single "VIP program."

Similarly, K-Means Cluster 3 (Long-Session Non-Converters: Avg Time ~101 min, Avg Purchases ~10.7) spreads across PAM 1 (275), PAM 3 (253), and PAM 4 (144). PAM's device lens separates these passive browsers into Desktop sub-groups (PAM 1 & 3) and a Mobile/Tablet fringe (PAM 4) — enabling more targeted friction-reduction campaigns per platform rather than a one-size-fits-all reactivation approach.

Strategic Conclusion — What Stability Tells Us:

Stability Level	K-Means	PAM	Count	Persona Label
Most Stable	4	3	354	Desktop Mid-Tier Transactor
Most Stable	1	4	353	Mobile Dormant User
Moderately Stable	2	2	340	Mobile Power Spender
Moderately Stable	3	1	275	Desktop Long-Session Browser
Fragmented	2	1/3	192/290	Split Power Users (Device-Driven)

The two stable personas — Desktop Mid-Tier Transactors and Mobile Dormant Users — should be treated as **production-ready segments** immediately deployable in CRM and campaign systems. The fragmented Power User zone, while numerically significant, requires the PAM categorical layer to be operationally useful, validating the necessity of the multi-method approach undertaken in this report.

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7 Strategic Conclusions

This analysis moves beyond surface-level behavioral metrics to deliver a **multidimensional, algorithmically validated understanding of user identity**. By triangulating findings across K-Means, K-Modes, and PAM, we have established that no single method is sufficient — behavioral intensity, categorical identity, and unified persona profiles each reveal a distinct and non-redundant layer of user truth.

The four archetypes that emerge consistently across methods are:

- **The Desktop Mid-Tier Transactor** — a stable, organically acquired Premium-plan user with moderate purchase frequency and a strong affinity for Books and Electronics on Desktop; the most algorithmically robust segment (PAM 3 × K-Means 4: 354 users) and the highest-priority retention target.
- **The Mobile Power Spender** — a referral-acquired, Premium-plan user on Mobile with high purchase frequency (~505 avg. past purchases) and consistent platform engagement; the platform's primary revenue engine requiring mobile-first loyalty strategies and channel-specific VIP programming.
- **The Long-Session Non-Converter** — a high time investment user (~101 min avg. session) with near-zero transactional output (~10.7 avg. past purchases), browsing passively across a wide range of session lengths; the highest-potential conversion opportunity through friction-reduction, personalized first-purchase incentives, and browse- abandonment triggers.
- **The Dormant Casual** — a Paid-Ad acquired, Free-tier user with low engagement, near- zero purchases, and minimal platform interaction; predominantly Mobile with a low feedback rate (77.6% no feedback), making them a churn-risk segment requiring reactivation nudges or sunset evaluation.

The stability analysis confirms that the Desktop Mid-Tier Transactor and Mobile Dormant User are **production-ready segments** with high cross-algorithm consistency, while the Mobile Power Spender and Long-Session Non-Converter require the PAM categorical lens to be fully operationally actionable. Together, these profiles form a data-driven foundation for targeted marketing, tiered product development, and customer success prioritization.

Final Data Integrity: The consolidated analytical dataset — combining behavioral scores, categorical modes, and unified cluster assignments — has been exported as `consolidated_clustering_results.csv` for immediate deployment in production CRM and campaign management environments.

```r
write.csv(comparison_df, "consolidated_clustering_results.csv", row.names = FALSE)
```